Rahul Sajnani | rahul_sajnani@brown.edu | Homepage | Scholar | GitHub | LinkedIn

Education

Brown University, Ph.D. in Computer Science Sep 2022 – Dec 2026
Advisor: Prof. Srinath Sridhar • CGPA 4.0/4.0 *Providence, RI*

IIIT Hyderabad, B.Tech. & M.Sc. in ECE Aug 2017 – May 2022
Gold Medal, Dean's List • CGPA 9.1/10 *Hyderabad, India*

Research Experience

Amazon Robotics Jun 2024 – Jan 2026
Applied Scientist II Intern, North Reading, MA *Managers: Jeroen Vanbaar, Jose Barreiros, Manikantan Nambi*

- **UniWALM** — a unified world-action-language model for memory-dependent robot manipulation ("reasoning at the speed of action"). A Mixture-of-Transformers (Wan2.2 video backbone + action DiT + **discrete diffusion language expert**) co-generates future video, robot action, and a short natural-language memory in a **single parallel diffusion pass**; the memory is re-read by the policy on every control step, so the robot remembers what it just did. Real-world memory-dependent manipulation: **11.1**% → **61.1**% vs same-backbone FastWAM; **34.5**% on RMBench (beats Mem-0's privileged-keyframe setup at 28.5%).
- **GeoDiffuser** — a training-free 2D/3D image editor built on a frozen T2I diffusion model, released as WACV 2025 Best Student Paper (invited talk). Users specify a 3D rigid transformation and GeoDiffuser rewrites the attention maps to realize it with no fine-tuning.
- **Data & systems:** led the design of bimanual and memory-dependent data collection pipelines (Shadow Hand, Franka Panda, AgileX dual-arm in sim); authored the unified-WAM design docs adopted cross-team.

Adobe Research May 2025 – Aug 2025
Research Scientist Intern, San Jose, CA *Advisors: Matheus Gadelha, Radomír Měch*

- **AppearancePointers** — a FLUX-based model for multimodal region control in image generation. Each region of the output is independently steered by any mix of text, reference image, and material texture in a single denoising pass. State-of-the-art on all three multi-region fidelity metrics, beating DreamRenderer and MSDiffusion at substantially larger scale; curated the AP-37K multimodal dataset.

Brown Visual Computing, Brown University Sep 2022 – Present
Graduate Researcher with Srinath Sridhar & Leonidas Guibas *Providence, RI*

- **3D structure for perception and generation:** SO(3)-equivariant pose on partial point clouds (ConDor, CVPR 2022) and NeRFs (Cafi-Net, CVPR 2023 Highlight); regularity-aware scene rearrangement (LEGO-Net, CVPR 2023).
- Currently: feed-forward 3D Gaussian Splatting from a single monocular video by distilling video diffusion backbones (Wan2.1, LTX-Video, CogVideoX).

Robotics Research Center (RRC), IIIT Hyderabad Aug 2019 – May 2022
Undergraduate Researcher with Prof. K. Madhava Krishna *Hyderabad, India*

- **DRACO** — weakly-supervised dense pose canonicalization (*IEEE ICRA 2021*) [arXiv]; multi-object monocular SLAM for dynamic driving scenes (*IEEE IV 2020*); designed the full **self-driving perception stack** [Video].

Publications

UniWALM: Unified World-Action-Language Model for Embodied Robot Control. *Under Review* [Project Page] 2026
Rahul Sajnani, Diego Romeres, Mingyuan Zhang, Kai Gao, Carl Qi, Baichuan Huang, Srinath Sridhar, Dylan Randle, Lillian J. Ratliff, Jose Barreiros, Manikantan Nambi, Jeroen Vanbaar.

AppearancePointers: Fine-Grained Multi-Region Control with Appearance Pointers. *Under Review* [Project Page] 2026
Rahul Sajnani, Matheus Gadelha, Yulia Gryaditskaya, Srinath Sridhar, Radomír Měch.

GeoDiffuser: Geometry-Based Image Editing with Diffusion Models. *IEEE WACV 2025 (Best Student Paper, invited talk)* [Project Page] [Code] 2025
Rahul Sajnani, Jeroen Vanbaar, Jie Min, Kapil Katyal, Srinath Sridhar.

GenHSI: Training-Free Controllable Generation of Human-Scene Interaction Videos. *IEEE WACV 2026* [Project Page] [arXiv] 2026
Zekun Li, Rui Zhou, **Rahul Sajnani**, Xiaoyan Cong, Daniel Ritchie, Srinath Sridhar.

ConDor: Self-Supervised Canonicalization of 3D Pose for Partial Shapes. *IEEE CVPR 2022* [Project Page] [Code] 2022
Rahul Sajnani, Adrien Poulenard, Jivitesh Jain, Radhika Dua, Leonidas J. Guibas, Srinath Sridhar.

Canonical Fields: Self-Supervised Learning of Pose-Canonicalized Neural Fields. *IEEE CVPR 2023 (Highlight)* [Project Page] [Code] 2023
Rohith Agaram, Shaurya Dewan, **Rahul Sajnani**, Adrien Poulenard, Madhava Krishna, Srinath Sridhar.

LEGO-Net: Learning Regular Rearrangements of Objects in Rooms. *IEEE CVPR 2023* [Project Page] 2023
Qiuhong Anna Wei, Sijie Ding, Jeong Joon Park, **Rahul Sajnani**, Adrien Poulenard, Srinath Sridhar, Leonidas Guibas.

DRACO: Weakly Supervised Dense Reconstruction And Canonicalization of Objects. *IEEE ICRA 2021* [arXiv] 2021
Rahul Sajnani, AadilMehdi Sanchawala, Krishna Murthy Jatavallabhula, Srinath Sridhar, K. Madhava Krishna.

Awards & Service

WACV 2025 Best Student Paper (GeoDiffuser, invited talk at WACV). **Gold Medal**, IIIT Hyderabad (2022, highest standing in ECE). **CVPR 2025 Outstanding Reviewer**. Co-Instructor Topics in 3D CV & ML (CSCI 2952K), Brown (2025). Head Organizer Robotics Summer School, IIIT-H (2021). **Reviewer:** CVPR, ICCV, ECCV, ICLR, SIGGRAPH.